Mirror-lattice arithmetic and the residual geometric gap for H-polarised Tyurin degenerations of K3 surfaces

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Abstract

We analyse the mirror-pair existence problem for a very general H-polarised Tyurin degeneration of K3 surfaces and a very general $H \oplus E_8 \oplus E_8$-quasipolarised K3 surface. We prove the lattice-theoretic prerequisites in the case $L = H$: the orthogonal complement of L is $\mathrm{II}_{2,18}$, every primitive rank-two isotropic sublattice is doubly admissible, the mirror lattice is $H \oplus E_8 \oplus E_8$, the admissibility parameter is $m = 1$, and the relevant primitive embeddings are unique. We also record the conditional consequences of an actual mirror pair, the non-balanced Type II monodromy and q-arithmetic, a self-contained consistency check for the $V_6 \cup V_{12}$ ansatz, an explicit monodromy obstruction on an elliptic fibration with singular fibres $2\,\mathrm{II}^* + 4\,\mathrm{I}_1$ for most q-values, and the fact that the corrected positivity clause $(A|_{V_i})^2 > 0$ is not consumed by the mirror-side lattice and monodromy axioms. The original existential assertion is the very general $L = H$ instance of the forward direction of Giovenzana–Thompson Conjecture 5.10; the cited work labels this direction as conjectural and does not supply the elliptic fibration or the allowable loop. Constructing that compatible pair remains the geometric gap.

1 Introduction and status of the mirror-pair assertion

The problem concerns a Tyurin degeneration

$$\pi\colon \mathcal{X} \to \Delta$$

of K3 surfaces with central fibre $X_0 = V_1 \cup V_2$, which is H-polarised in the corrected sense. Thus, over the punctured disc Δ^* there is a trivial local subsystem $\mathcal{L} \subset R^2\pi_*\mathbb{Z}$ with fibre H, and there is an effective flat nef Cartier divisor A on $\mathcal{X}$ such that A_p is nef and big for $p \neq 0$, $[A_p] \in \mathcal{L}_p$, and

$$(A|_{V_i})^2 > 0, \qquad i = 1, 2.$$

The degeneration is assumed very general in the Alexeev–Brunyate–Engel compactified moduli sense [1]. The question asks whether, for a very general $H \oplus E_8 \oplus E_8$-quasipolarised K3 surface Y, there is an elliptic fibration $Y \to \mathbb{P}^1$ and an allowable loop γ compatible with the quasipolarisation, such that the two objects form a mirror pair in the sense of Giovenzana–Thompson [3]. The main conclusion of this paper is a precise status statement.

Theorem 1.1 (Status)**.** *The existential assertion in the original problem is the very general $L = H$ instance of the forward direction of Giovenzana–Thompson Conjecture 5.10 [3], after identifying the mirror lattice $\check{L}$ with $H \oplus E_8 \oplus E_8$. Giovenzana–Thompson state this forward direction as a conjecture and do not supply the elliptic fibration $\pi\colon Y \to \mathbb{P}^1$ or the allowable loop γ.*

The lattice-theoretic prerequisites for this instance are proved in Theorem 3.1 and Proposition 3.2; the closed lattice-incidence reduction is Proposition 2.3; conditional consequences of an actual mirror pair are Theorem 4.1; and the non-balanced monodromy arithmetic is Theorem 5.1. The construction of an elliptic fibration $\pi\colon Y \to \mathbb{P}^1$ and an allowable loop γ satisfying Definition 4.21 compatibility is therefore the remaining geometric obligation.

Proof. Giovenzana–Thompson Conjecture 5.10 [3] predicts, in its forward direction, that an L-polarised Tyurin degeneration with a suitable m-admissible primitive isotropic class in $L^\perp$ gives rise to a $\check{L}$-quasipolarised K3 surface with an elliptic fibration and an allowable loop, compatible in the sense of [3, Definition 4.21], and forming a mirror pair in the sense of [3, Definition 5.1]. The introduction and Remark 5.11 of [3] identify the choice of the loop γ, and conversely the choice of birational model, as the difficult part, not determined by lattice theory alone.

For $L = H$, Theorem 3.1 below proves that

$$L^\perp \cong H \oplus H \oplus E_8 \oplus E_8 \cong \mathrm{II}_{2,18},$$

that every primitive rank-two isotropic sublattice $I \subset L^\perp$ is doubly admissible, and that for a primitive isotropic generator $e_1 \in I$ the mirror lattice

$$\check{L} = {e_1}^{\perp}_{L^\perp}/\mathbb{Z}e_1$$

is even unimodular of signature $(1, 17)$, hence isomorphic to $H \oplus E_8 \oplus E_8$. The same theorem proves that the relevant admissibility clauses hold with $m = 1$. Therefore the quasipolarisation lattice appearing in the original problem is exactly the mirror lattice predicted by the $L = H$ case of the conjecture.

The very general hypothesis in the problem excludes Noether–Lefschetz loci in the compactified moduli space of [1]; it does not supply the fibration or the loop. The monodromy-invariance clause is already part of the definition of an H-polarised Tyurin degeneration: the local subsystem $\mathcal{L}$ is required to be trivial. Consequently, the original existential statement is precisely the very general $L = H$ case of the forward mirror construction conjectured in [3], and the part not supplied by the cited lattice criteria is the existence of the compatible pair (π, γ). □

The rest of the paper proves the lattice, incidence, monodromy and obstruction statements used in this status assessment, and isolates the remaining geometric obligations.

2 Notation and the Giovenzana–Thompson criteria

Convention 2.1. All lattices are free $\mathbb{Z}$-modules equipped with integral non-degenerate symmetric bilinear forms. The hyperbolic plane H has a basis u, v with $u^2 = v^2 = 0$ and

$\langle u, v\rangle = 1$. The lattice E_8 is the negative definite even unimodular E_8 root lattice. The K3 lattice is

$$\Lambda_{\mathrm{K3}} = H^{\oplus 3} \oplus E_8^{\oplus 2},$$

an even unimodular lattice of signature $(3, 19)$. For a primitive sublattice $L \subset \Lambda_{\mathrm{K3}}$, we write $L^\perp$ for its orthogonal complement in Λ_{K3}. If $I \subset L^\perp$ is a primitive rank-two isotropic sublattice, we write $I^\perp/I$ for the quotient of the orthogonal complement of I in Λ_{K3} by I.

We write $\mathrm{II}_{r,s}$ for the unique indefinite even unimodular lattice of signature (r, s) when $r-s \equiv 0 \pmod 8$. For a surface or fibration piece W, the symbol $\mathrm{e}(W)$ denotes its topological Euler number.

We use the admissibility terminology of [3].

Definition 2.2 ([3, Definitions 3.30 and 3.31])**.** Let M be a non-degenerate lattice and $e \in M$ a primitive isotropic vector. The divisibility $\mathrm{div}(e)$ is the positive generator of the image of the map

$$\langle e, -\rangle_M \colon M \to \mathbb{Z}.$$

The vector e is m-admissible in M if $\mathrm{div}(e) = m$ and there is an isotropic class $g \in M$ with $\mathrm{div}(g) = m$ and $\langle e, g\rangle = m$.

A primitive rank-two isotropic sublattice $I = \langle e_1, e_2\rangle \subset M$ is doubly admissible in M if there exist isotropic classes $g_1, g_2 \in M$ such that

$$\langle e_1, g_1\rangle = \langle e_2, g_2\rangle = 1, \qquad \langle e_1, g_2\rangle = \langle e_2, g_1\rangle = 0.$$

An L-polarised Tyurin degeneration is doubly admissible if the associated monodromy lattice $I \subset L^\perp$ is doubly admissible in $L^\perp$.

We use the following criterion from [3].

Proposition 2.3 (Giovenzana–Thompson criterion)**.** *Assume that an L-polarised Tyurin degeneration $X \to \Delta$ and an $\check{L}$-quasipolarised K3 fibration Y are both doubly admissible. Then they form a mirror pair in the sense of [3, Definition 5.1] if and only if there exists an isometry*

$$\psi \colon (\mathbb{Z}F \oplus \mathbb{Z}\tau)^\perp/(\mathbb{Z}F \oplus \mathbb{Z}\tau) \longrightarrow I^\perp/I$$

such that

$$\psi(\Gamma) = (\mathrm{im}\, L)^\perp_{I^\perp/I}$$

and, for $i = 1, 2$,

$$\psi \circ \phi_Y\big([r_i(b)]^\perp\big) = \phi_X\big(K_{V_i}^\perp\big).$$

Here F is the fibre class on the mirror side, τ is the lifted class in $\check{L}^\perp$, Γ is the natural rank-16 negative-definite quotient lattice on the mirror side, and ϕ_X, ϕ_Y are the period-lattice identifications used in [3].

In the source [3, Proposition 5.3], the first displayed condition is written as $\psi(\Gamma) = L^\perp$ inside $I^\perp/I$. This means the orthogonal complement of the image of L in the quotient $I^\perp/I$: the form on $I^\perp$ descends to $I^\perp/I$ because I is isotropic, and L maps to a primitive sublattice. In the case $L = H$, this complement is identified with Γ in Theorem 3.1.

Proof. This is precisely [3, Proposition 5.3]. The only hypotheses used there are the double admissibility of both sides. In the forward direction the isometry ψ is part of Definition 5.1. In the converse direction, double admissibility supplies the primitivity clauses required by Definition 5.1(1) via [3, Propositions 3.33 and 4.28]; the lifting clause Definition 5.1(3) is obtained by the argument of [3, Lemma 3.32], since the relevant primitive isotropic vectors lie in hyperbolic-plane summands and are unique up to isometry; and the m-admissibility clause Definition 5.1(4) follows from the admissibility data. Thus the existence of such a ψ is equivalent to the mirror-pair condition under the stated hypotheses. □

Proposition 2.4 (Marking invariance)**.** *Conditions (1)–(3) of [3, Definition 5.1] depend only on the isometry classes of the lattice data*

$$(L,\ I \subset L^\perp,\ \Gamma,\ \psi),$$

not on the choice of marking. The only choice-dependence is the labelling of V_1 and V_2, and the global sign ambiguity already present in the pseudolattice comparison.

Proof. The conditions in Definition 5.1(1)–(3) are formulated in terms of primitive embeddings, quotient lattices, and the existence of an isometry ψ between quotient lattices satisfying subspace equalities. None of these clauses refers to a basis or to a particular marking.

A change of marking conjugates the period identifications by isometries of the ambient K3 lattice. Such isometries induce isometries of the quotient lattices

$$I^\perp/I \quad \text{and} \quad (\mathbb{Z}F \oplus \mathbb{Z}\tau)^\perp/(\mathbb{Z}F \oplus \mathbb{Z}\tau).$$

If an isometry ψ satisfying the required equalities exists before the marking change, then conjugating it by the induced isometries gives one after the marking change, and conversely. The equalities involving $K_{V_i}^\perp$ and $[r_i(b)]^\perp$ are preserved under these induced isometries.

Swapping the labels V_1 and V_2 swaps the two indexed equalities, and the global sign ambiguity of [3, Proposition 2.37] preserves all inner products. No further dependence on markings remains. □

3 The mirror lattice for $L = H$

We now prove the explicit lattice arithmetic for the case $L = H$.

Theorem 3.1. *Let $L \cong H$ be primitively embedded in Λ_{K3}, and let $I = \langle e_1, e_2\rangle \subset L^\perp$ be a primitive rank-two isotropic sublattice with primitive isotropic generators e_1, e_2. Then:*

1. *$L^\perp \cong H \oplus H \oplus E_8 \oplus E_8 \cong \mathrm{II}_{2,18}$.*
2. *I is doubly admissible in $L^\perp$.*
3. *With respect to the double admissibility data, there is an orthogonal decomposition*

$$L^\perp = H_1 \oplus H_2 \oplus \Gamma,$$

where $H_1 = \langle e_1, g_1\rangle \cong H$, $H_2 = \langle e_2, g_2\rangle \cong H$, and Γ is an even unimodular negative definite lattice of rank 16.

4. *The mirror lattice*
$$\check{L} = e_1{}^{\perp}_{L^{\perp}}/\mathbb{Z}e_1$$
is isomorphic to $H_2 \oplus \Gamma$*, hence is even unimodular of signature* $(1, 17)$ *and therefore*
$$\check{L} \cong \mathrm{II}_{1,17} \cong H \oplus E_8 \oplus E_8.$$

5. $\mathrm{div}(e_1) = \mathrm{div}(e_2) = 1$ *in* $L^{\perp}$*, and the* m*-admissibility clauses hold with* $m = 1$*. If the mirror-side complement* $\check{L}^{\perp}$ *is identified with* $H \oplus H$ *and* τ *is chosen as the primitive isotropic vector corresponding to* e_1 *under the lift used in Definition 5.1(3), then* τ *is also* 1*-admissible.*

6. *One has*
$$I^{\perp}/I \cong L \oplus \Gamma \cong H \oplus \Gamma \cong H \oplus E_8 \oplus E_8,$$
and the orthogonal complement of the image of L *in* $I^{\perp}/I$ *is naturally isomorphic to* Γ*. The embedding* $L \hookrightarrow I^{\perp}/I$ *is primitive.*

Proof. Since $L \cong H$ is unimodular of signature $(1, 1)$ and Λ_{K3} is even unimodular of signature $(3, 19)$, the primitive unimodular sublattice L is an orthogonal direct summand:
$$\Lambda_{\mathrm{K3}} = L \oplus L^{\perp}.$$
Indeed, for $x \in \Lambda_{\mathrm{K3}}$, the functional $\ell \mapsto \langle x, \ell \rangle$ on L is represented by a unique element of L because L is unimodular; subtracting this element leaves a vector orthogonal to L. Hence $L^{\perp}$ is even unimodular of signature
$$(3, 19) - (1, 1) = (2, 18)$$
and rank 20. By the classification of indefinite even unimodular lattices [5], there is a unique such lattice:
$$L^{\perp} \cong \mathrm{II}_{2,18} \cong H \oplus H \oplus E_8 \oplus E_8.$$

We now prove double admissibility. Because $L^{\perp}$ is unimodular and I is primitive, the pairing map
$$L^{\perp} \longrightarrow \mathrm{Hom}(I, \mathbb{Z})$$
is surjective. Thus we may choose $h_1 \in L^{\perp}$ with
$$\langle e_1, h_1 \rangle = 1, \qquad \langle e_2, h_1 \rangle = 0.$$
Since $L^{\perp}$ is even, h_1^2 is even. Put
$$g_1 = h_1 - \frac{h_1^2}{2} e_1.$$
Then
$$g_1^2 = h_1^2 - h_1^2 \langle e_1, h_1 \rangle = 0,$$
and the pairings with e_1, e_2 are unchanged, so
$$\langle e_1, g_1 \rangle = 1, \qquad \langle e_2, g_1 \rangle = 0.$$

Similarly choose h_2 with

$$\langle e_2, h_2\rangle = 1, \qquad \langle e_1, h_2\rangle = 0.$$

Replace h_2 by

$$h_2' = h_2 - \langle g_1, h_2\rangle e_1,$$

so that $\langle g_1, h_2'\rangle = 0$, while the pairings with e_1, e_2 remain unchanged. Since h_2' has even square, put

$$g_2 = h_2' - \frac{(h_2')^2}{2} e_2.$$

Then g_2 is isotropic and satisfies

$$\langle e_2, g_2\rangle = 1, \qquad \langle e_1, g_2\rangle = 0, \qquad \langle g_1, g_2\rangle = 0.$$

Thus g_1, g_2 are the required double-admissibility witnesses. The pairs $H_1 = \langle e_1, g_1\rangle$ and $H_2 = \langle e_2, g_2\rangle$ are hyperbolic planes, mutually orthogonal by construction. Let

$$\Gamma = (H_1 \oplus H_2)^{\perp}_{L^\perp}.$$

Because $H_1 \oplus H_2$ is unimodular and $L^\perp$ is unimodular, Γ is an orthogonal direct summand and is unimodular. It is even because $L^\perp$ is even. Its rank is $20 - 4 = 16$, and its signature is

$$(2, 18) - (2, 2) = (0, 16),$$

so Γ is negative definite. The precise isomorphism class of Γ is not needed; the argument below uses only that it is even, unimodular and negative definite.

Inside $H_1 = \langle e_1, g_1\rangle$, the vectors orthogonal to e_1 are exactly the multiples of e_1, since

$$\langle a e_1 + b g_1, e_1\rangle = b.$$

Therefore

$$e_1{}^{\perp}_{L^\perp} = \mathbb{Z}e_1 \oplus H_2 \oplus \Gamma,$$

and quotienting by $\mathbb{Z}e_1$ gives

$$\check{L} = e_1{}^{\perp}_{L^\perp}/\mathbb{Z}e_1 \cong H_2 \oplus \Gamma.$$

This lattice is even (quotient of an even lattice by an isotropic vector). It is unimodular because in the above decomposition it is the direct summand $H_2 \oplus \Gamma$ of the unimodular lattice $L^\perp$. Its signature is $(1, 17)$. By the classification of indefinite even unimodular lattices [5],

$$\check{L} \cong \mathrm{II}_{1,17} \cong H \oplus E_8 \oplus E_8.$$

Since $\langle e_1, g_1\rangle = 1$, the divisibility of e_1 in $L^\perp$ is 1; similarly $\mathrm{div}(e_2) = 1$. The isotropic witness g_1 has divisibility 1 because $\langle e_1, g_1\rangle = 1$, and likewise for g_2. Hence e_1 and e_2 are 1-admissible in $L^\perp$. The mirror-side complement $\check{L}^\perp$ is also even unimodular, of signature $(2, 2)$, hence isomorphic to $H \oplus H$. After choosing the standard lift identifying τ with a primitive isotropic vector in one H-summand, the same argument gives $\mathrm{div}(\tau) = 1$, so the mirror-side m-admissibility clause also holds with $m = 1$.

Finally, in the decomposition

$$L^\perp = H_1 \oplus H_2 \oplus \Gamma,$$

the orthogonal complement of $I = \langle e_1, e_2 \rangle$ inside $L^\perp$ is

$$I^\perp \cap L^\perp = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \Gamma = I \oplus \Gamma.$$

Inside Λ_{K3}, since L is orthogonal to $L^\perp$, we have

$$I^\perp_{\Lambda_{\mathrm{K3}}} = L \oplus I \oplus \Gamma.$$

Quotienting by I gives

$$I^\perp / I \cong L \oplus \Gamma.$$

Because $L \cong H$ and Γ is negative definite unimodular of rank 16, the indefinite lattice $L \oplus \Gamma$ has signature $(1, 17)$ and is even unimodular, hence isomorphic to $H \oplus E_8 \oplus E_8$ by the indefinite classification. The image of L is an orthogonal direct summand, so the embedding $L \hookrightarrow I^\perp / I$ is primitive, and its orthogonal complement is Γ. □

The primitive embedding of the mirror lattice into the K3 lattice is also rigid.

Proposition 3.2. *Let*

$$M = H \oplus E_8 \oplus E_8.$$

There is, up to isometry of Λ_{K3}, a unique primitive embedding

$$M \hookrightarrow \Lambda_{\mathrm{K3}},$$

and its orthogonal complement is isomorphic to $H \oplus H$. Moreover, in the decomposition of Theorem 3.1, the primitive embedding $L \hookrightarrow I^\perp_{\Lambda_{\mathrm{K3}}}$ factors through primitive embeddings

$$L \hookrightarrow M' \cong M \hookrightarrow I^\perp_{\Lambda_{\mathrm{K3}}},$$

where $M' = L \oplus \Gamma$.

Proof. The lattice M is even unimodular of signature $(1, 17)$, and Λ_{K3} is even unimodular of signature $(3, 19)$. In the standard decomposition

$$\Lambda_{\mathrm{K3}} = (H \oplus H) \oplus (H \oplus E_8 \oplus E_8),$$

the second summand gives an explicit primitive embedding of M with complement $H \oplus H$.

For uniqueness, we invoke Nikulin's primitive embedding theorem [6, Proposition 1.6.1]. Its hypotheses are satisfied because both M and the desired complement $H \oplus H$ are indefinite even lattices with trivial discriminant groups, and the rank and signature conditions are

$$\operatorname{rank} M = 18, \qquad \operatorname{rank}(H \oplus H) = 4, \qquad (1, 17) + (2, 2) = (3, 19).$$

The discriminant-form obstruction is therefore vacuous. Hence the primitive embedding is unique up to isometry of Λ_{K3}, and the complement is forced to be $H \oplus H$.

For the factorisation, use the decomposition from Theorem 3.1:

$$I_{\Lambda_{\mathrm{K3}}}^{\perp} = L \oplus I \oplus \Gamma.$$

Let

$$M' = L \oplus \Gamma.$$

Then $M' \cong H \oplus \Gamma$ is even unimodular of signature $(1, 17)$, hence $M' \cong M$ by the indefinite classification [5]. The inclusion $M' \hookrightarrow I_{\Lambda_{\mathrm{K3}}}^{\perp}$ is primitive because the quotient is the free lattice I, and $L \hookrightarrow M'$ is primitive because it is an orthogonal direct-summand inclusion. Thus

$$L \hookrightarrow M' \cong M \hookrightarrow I_{\Lambda_{\mathrm{K3}}}^{\perp}.$$

□

4 Consequences conditional on the existence of a mirror pair

We now record what follows if a mirror pair actually exists.

Theorem 4.1. *Assume that an H-polarised Tyurin degeneration $X_0 = V_1 \cup V_2$ and an $H \oplus E_8 \oplus E_8$-quasipolarised K3 fibration form a mirror pair in the sense of [3, Definition 5.1]. Then for each $j = 1, 2$ there are pseudolattice isomorphisms*

$$H_2(Y_j, F_p; \mathbb{Z}) \cong K_0^{\mathrm{num}}(D(V_j))$$

and

$$H_1(F_p; \mathbb{Z}) \cong K_0^{\mathrm{num}}(D(C)),$$

making the natural square commute. Moreover, after fixing the auxiliary chamber/polarization datum h required in [2, Theorem 4.7], the corresponding chambered analytic moduli spaces satisfy

$$\dim \overline{\mathcal{M}}_{H,h} = 18, \qquad \dim \overline{\mathcal{M}}_{H \oplus E_8 \oplus E_8, h} = 2.$$

Proof. The pseudolattice statement is [3, Theorem 5.4]; its hypotheses are exactly that the two objects form a mirror pair in the sense of Definition 5.1. Its proof uses Definition 5.1(2) together with [3, Proposition 2.37] to lift the mirror isometry to isometries

$$\psi_j \colon \mathrm{NS}(H_2(Y_j, F_p; \mathbb{Z})) \to \mathrm{NS}(V_j)$$

satisfying

$$\psi_j([r_j(b)]) = -K_{V_j},$$

and then applies [3, Proposition 2.21] to obtain the pseudolattice isomorphisms.

For the dimension statement, we use Theorem 4.7 of Alexeev–Engel [2]. That theorem concerns the chambered moduli space $\overline{\mathcal{M}}_{\Lambda,h}$ associated with a non-degenerate lattice Λ and an auxiliary chamber/polarization datum h. For such a datum, the theorem gives a smooth complex analytic space of dimension

$$20 - \operatorname{rank} \Lambda.$$

The lattices in question here are non-degenerate: H has signature $(1,1)$, and $H \oplus E_8 \oplus E_8$ has signature $(1,17)$. For $L = H$, $\operatorname{rank} H = 2$, so the H-polarised side has dimension

$$20 - 2 = 18.$$

For the mirror side, $\operatorname{rank}(H \oplus E_8 \oplus E_8) = 18$, so the dimension is

$$20 - 18 = 2.$$

□

5 Non-balanced Type II monodromy and the $V_6 \cup V_{12}$ consistency check

We state the Type II monodromy filtration only under an explicit non-balanced hypothesis. The balanced case $q = 0$ is left open.

Theorem 5.1. *Let $\pi\colon \mathcal{X} \to \Delta$ be an H-polarised Tyurin degeneration of K3 surfaces with central fibre $X_0 = V_1 \cup_C V_2$, and let X be a nearby smooth fibre. Let T be the local monodromy on $H^2(X;\mathbb{Z})$ and $N = \log T$.*

1. *Assume explicitly that the degeneration is non-balanced Type II in the sense that T is unipotent, $N^2 = 0$, and $I = \operatorname{im} N$ is a primitive rank-two isotropic sublattice. Then $I \subset L^\perp$,*

   $$I^\perp = \ker N,$$

   and the filtration

   $$0 \subset I = \operatorname{im} N \subset I^\perp = \ker N \subset H^2(X;\mathbb{Z})$$

   is the relative monodromy weight filtration in the sense of the Friedman–Scattone theory as summarized in [3, §3.1]. Moreover,

   $$I^\perp / I \cong H \oplus E_8 \oplus E_8.$$

   No assertion is made here for the balanced case $q = 0$.

2. *On a candidate mirror fibration split by a loop γ, impose the mirror-pairing relation*

   $$12 - \mathrm{e}(Y_1) = K_{V_1}^2 =: q$$

   and assume $q \neq 0$. If Assumption 4.9(1) of [3] holds for the first disc fibration, then the anticlockwise boundary monodromy matrices are

   $$T_1 = \begin{pmatrix} 1 & -q \\ 0 & 1 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 1 & q \\ 0 & 1 \end{pmatrix},$$

   and the Euler numbers satisfy

   $$\mathrm{e}(Y_1) = 12 - q, \qquad \mathrm{e}(Y_2) = 12 + q.$$

Allowability of the loop γ in the sense of [3, Definition 4.11] requires both clauses of Assumption 4.9. The first clause is exactly the displayed matrix condition; the second clause, primitivity of $r(a)$ in the relevant relative homology, is retained as an explicit hypothesis and is not proved here.

Proof. For the first part, the explicit non-balanced Type II hypothesis gives T unipotent, $N^2 = 0$, and $I = \operatorname{im} N$ primitive rank two. Since T is an isometry of the intersection form, its logarithm N is infinitesimally orthogonal:

$$\langle Nx, y\rangle + \langle x, Ny\rangle = 0.$$

Because the H-polarisation is monodromy invariant, T acts trivially on L, hence N acts trivially on L. Thus, for $x \in H^2(X;\mathbb{Z})$ and $\ell \in L$,

$$\langle Nx, \ell\rangle = -\langle x, N\ell\rangle = 0,$$

so $I = \operatorname{im} N \subset L^\perp$. The isotropy of I follows from $N^2 = 0$ and infinitesimal orthogonality:

$$\langle Nx, Ny\rangle = -\langle x, N^2 y\rangle = 0.$$

We now derive the equality $I^\perp = \ker N$. For $y \in H^2(X;\mathbb{Z})$,

$$y \in (\operatorname{im} N)^\perp \iff \langle Nx, y\rangle = 0 \quad \text{for all } x$$

$$\iff -\langle x, Ny\rangle = 0 \quad \text{for all } x$$

by infinitesimal orthogonality. Since the intersection form on $H^2(X;\mathbb{Z})$ is non-degenerate, this is equivalent to $Ny = 0$. Hence

$$I^\perp = (\operatorname{im} N)^\perp = \ker N.$$

For a nilpotent operator N with $N^2 = 0$, the relative monodromy weight filtration of the limiting mixed Hodge structure is the filtration whose nontrivial steps are $\operatorname{im} N$ and $\ker N$; in the present Type II K3 setting this is the Friedman–Scattone filtration as summarized in [3, §3.1]. Thus the displayed filtration is precisely that weight filtration.

Finally, since $I \subset L^\perp$ is a primitive rank-two isotropic sublattice, Theorem 3.1 gives

$$I^\perp / I \cong H \oplus E_8 \oplus E_8.$$

For the second part, Assumption 4.9(1) of [3] gives, in a symplectic basis of $H_1(F;\mathbb{Z})$,

$$T_1 = \begin{pmatrix} 1 & \mathrm{e}(Y_1) - 12 \\ 0 & 1 \end{pmatrix}.$$

Using $q = 12 - \mathrm{e}(Y_1)$ gives

$$T_1 = \begin{pmatrix} 1 & -q \\ 0 & 1 \end{pmatrix}.$$

By [3, Remark 4.12], the second boundary monodromy is inverse to the first:

$$T_2 = T_1^{-1} = \begin{pmatrix} 1 & q \\ 0 & 1 \end{pmatrix}.$$

The same remark gives

$$\mathrm{e}(Y_1) + \mathrm{e}(Y_2) = 24,$$

so with $\mathrm{e}(Y_1) = 12 - q$ one obtains $\mathrm{e}(Y_2) = 12 + q$.

Finally, [3, Definition 4.11] declares γ allowable when both disc fibrations satisfy Assumption 4.9. Clause (1) is the matrix condition just verified. Clause (2), the primitivity of $r(a)$, is not a consequence of the Euler-number arithmetic and is therefore kept as an explicit hypothesis. □

We now check the self-contained numerical consistency of the $V_6 \cup V_{12}$ ansatz. The monodromy matrices below are derived directly from [3, Assumption 4.9(1) and Remark 4.12], not from the conditional Type II theorem above.

Proposition 5.2. *Let $C \subset \mathbb{P}^2$ be a smooth plane cubic and let $V_d = \mathrm{Bl}_d \mathbb{P}^2$ be the blow-up of $\mathbb{P}^2$ in d distinct points on C. Let $D \subset V_d$ be the strict transform of C. For the ansatz*

$$X_0 = V_6 \cup_D V_{12},$$

the following hold.

1. *On V_d one has*
$$D = -K_{V_d}, \qquad D^2 = K_{V_d}^2 = 9 - d.$$
Thus $\deg N_{D/V_6} = +3$ and $\deg N_{D/V_{12}} = -3$; the numerical d-semistability condition is satisfied, equivalently $6 + 12 = 18$.

2. *The topological Euler numbers are*
$$\mathrm{e}(V_6) = 9, \qquad \mathrm{e}(V_{12}) = 15,$$
and hence
$$\mathrm{e}(V_6) + \mathrm{e}(V_{12}) = 24.$$

3. *If disc fibrations Y_i with $\mathrm{e}(Y_i) = \mathrm{e}(V_i)$ satisfy Assumption 4.9(1) of [3], then their anticlockwise boundary monodromies are*
$$\begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix},$$
which are mutually inverse, and
$$12 - \mathrm{e}(Y_1) = 3 = K_{V_6}^2.$$

No existence, dominance, very-general occurrence, or allowability assertion is made.

Proof. Let h denote the pullback of the hyperplane class on $\mathbb{P}^2$, and let $E_1, \dots, E_d$ be the exceptional divisors over distinct points of C. Then

$$K_{V_d} = -3h + \sum_{i=1}^{d} E_i.$$

The strict transform of the cubic C has class

$$D = 3h - \sum_{i=1}^{d} E_i = -K_{V_d}.$$

Therefore

$$D^2 = K_{V_d}^2 = 9 - d.$$

For $d = 6$ this gives $D^2 = 3$, and for $d = 12$ it gives $D^2 = -3$. Since $\deg N_{D/V_d} = D^2$, the normal bundle degrees are $+3$ and -3, and their sum is 0. Thus the numerical condition for d-semistability is satisfied.

The Euler characteristic of $\mathbb{P}^2$ is 3, and each blow-up at a distinct point adds 1, so

$$\mathrm{e}(V_d) = 3 + d.$$

Hence

$$\mathrm{e}(V_6) = 9, \qquad \mathrm{e}(V_{12}) = 15,$$

and the sum is 24.

If Y_1 has Euler number 9 and satisfies Assumption 4.9(1) of [3], then by that assumption itself its anticlockwise boundary monodromy is

$$\begin{pmatrix} 1 & 9-12 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}.$$

For Y_2 with Euler number 15 one obtains

$$\begin{pmatrix} 1 & 15-12 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}.$$

These matrices are inverse, as required by [3, Remark 4.12]. Finally,

$$12 - \mathrm{e}(Y_1) = 12 - 9 = 3 = K_{V_6}^2.$$

□

6 Extremal-fibration obstruction and the corrected positivity clause

We now prove a monodromy obstruction for certain allowable-loop candidates on an elliptic fibration with singular fibres $2\,\mathrm{II}^* + 4\,\mathrm{I}_1$.

Theorem 6.1. *Let $\pi\colon Y \to \mathbb{P}^1$ be an elliptic fibration on a K3 surface whose singular fibres are*

$$2\,\mathrm{II}^* + 4\,\mathrm{I}_1,$$

with Euler numbers $\mathrm{e}(\mathrm{II}^*) = 10$ and $\mathrm{e}(\mathrm{I}_1) = 1$. Let γ be a simple loop enclosing a subset S of singular fibres, and put

$$\mathrm{e}_1 = \sum_{F \subset S} \mathrm{e}(F), \qquad q = 12 - \mathrm{e}_1\,.$$

Then:

1. *Assumption 4.9(1) of [3] holds for γ if and only if it holds for one of the two sides, since the two anticlockwise boundary monodromies are inverse and $\mathrm{e}_1 + \mathrm{e}_2 = 24$.*
2. *If*
$$|q| \in \{2, 3, 4, 5, 6, 7\} \quad \text{or} \quad |q| \geq 10,$$
then no such loop satisfies Assumption 4.9(1).
3. *The residual cases*
$$|q| \in \{0, 1, 8, 9\}$$
are not decided by the fibre configuration alone; they depend on the actual monodromy representation and vanishing-cycle conjugation data.
4. *In particular, the value $q = 3$ required by the $V_6 \cup V_{12}$ ansatz of Proposition 5.2 is excluded for this extremal fibration.*

Proof. The Euler numbers of the singular fibres are

$$\mathrm{e}(\mathrm{II}^*) = 10, \qquad \mathrm{e}(\mathrm{I}_1) = 1,$$

by Kodaira's table [4]. The multiset of Euler numbers is therefore

$$\{10, 10, 1, 1, 1, 1\}.$$

The possible subset sums are

$$0, 1, 2, 3, 4,$$

using only I_1 fibres;

$$10, 11, 12, 13, 14,$$

using exactly one II^* fibre; and

$$20, 21, 22, 23, 24,$$

using both II^* fibres. There are no subset sums $5, 6, 7, 8, 9, 15, 16, 17, 18, 19$.

Swapping the two sides replaces e_1 by $24 - \mathrm{e}_1$ and q by $-q$. Also, if one side has monodromy

$$\begin{pmatrix} 1 & \mathrm{e}_1 - 12 \\ 0 & 1 \end{pmatrix},$$

then the other side has the inverse matrix. Hence it suffices to consider $\mathrm{e}_1 \leq 12$, namely

$$\mathrm{e}_1 \in \{0, 1, 2, 3, 4, 10, 11, 12\}.$$

The corresponding q-values are

$$12, 11, 10, 9, 8, 2, 1, 0.$$

For the monodromy arguments, let

$$U_n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}.$$

For $A \in \mathrm{SL}_2(\mathbb{Z})$ define

$$c(A) = \gcd\{\text{entries of } A - I\},$$

with $c(I) = 0$. The integer $c(A)$ is invariant under $\mathrm{SL}_2(\mathbb{Z})$-conjugacy: if $B = PAP^{-1}$, then $B - I = P(A - I)P^{-1}$, and multiplication on the left and right by unimodular integral matrices preserves the ideal of $\mathbb{Z}$ generated by the entries. If A is conjugate to U_n, then $c(A) = |n|$. Assumption 4.9(1) requires the disc monodromy to be conjugate to $U_{\mathrm{e}_1 - 12}$, so it requires $c(A) = |\,\mathrm{e}_1 - 12| = |q|$.

We first exclude $q = 2$, i.e. $\mathrm{e}_1 = 10$. A subset with Euler sum 10 must consist of a single II^* fibre. The local monodromy of a II^* fibre has finite order 6 [4]; in particular it is not unipotent. It therefore cannot be conjugate to U_{-2}.

Next consider $q = 12$, i.e. $\mathrm{e}_1 = 0$. The enclosed subset is empty, so the monodromy is trivial, whereas Assumption 4.9(1) would require U_{-12}, which has $c = 12$. This is impossible.

For $q = 11$, i.e. $\mathrm{e}_1 = 1$, the enclosed subset is a single I_1 fibre. The monodromy is conjugate to U_1 up to orientation, so $c = 1$, whereas the required matrix has $c = 11$. Thus this case is impossible.

For $q = 10$, i.e. $\mathrm{e}_1 = 2$, the enclosed subset consists of two I_1 fibres. The disc monodromy is a product of two positive Dehn twists about primitive vanishing cycles. Write a positive Dehn twist about a primitive cycle v as

$$\tau_v(x) = x + (x \cdot v)v.$$

For two primitive cycles v, w on an elliptic curve, one computes

$$\mathrm{tr}(\tau_v \tau_w) = 2 - (v \cdot w)^2.$$

If the product is conjugate to a matrix U_n, its trace is 2, hence $v \cdot w = 0$. In rank two with a symplectic intersection form, primitive vectors with zero intersection are proportional, so $w = \pm v$. Since $\tau_{-v} = \tau_v$, the product is τ_v^2, hence conjugate to U_2, with $c = 2$. The required matrix has $c = 10$, so this case is impossible.

The values $|q| = 3, 4, 5, 6, 7$ would require $\mathrm{e}_1 = 9, 8, 7, 6, 5$ or the complementary values $15, 16, 17, 18, 19$, none of which is a subset sum of $\{10, 10, 1, 1, 1, 1\}$. The values $|q| \geq 13$ are impossible because $0 \leq \mathrm{e}_1 \leq 24$. Together with the explicit exclusions of $q = 10, 11, 12$, this proves the stated exclusion range.

The remaining cases are

$$\mathrm{e}_1 = 3, 4, 11, 12,$$

corresponding to

$$|q| = 9, 8, 1, 0.$$

For $\mathrm{e}_1 = 3, 4$ the disc monodromy is a product of three or four I_1 twists. For $\mathrm{e}_1 = 11, 12$ it involves a II^* monodromy multiplied by products of I_1 twists. In all these cases the trace and the gluing number depend on the conjugating matrices in the global monodromy representation

$$\pi_1(\mathbb{P}^1 \setminus \Sigma) \to \mathrm{SL}_2(\mathbb{Z}),$$

and are not determined solely by the fibre configuration. Hence these residual cases are left undecided.

Finally, the $V_6 \cup V_{12}$ ansatz of Proposition 5.2 has $q = 3$. Since $|q| = 3$ lies in the excluded range, the extremal fibration $2\,\mathrm{II}^* + 4\,\mathrm{I}_1$ cannot carry the required allowable loop for that value of q. □

We now analyse the corrected positivity condition

$$(A|_{V_i})^2 > 0.$$

Proposition 6.2. *The corrected positivity clause $(A|_{V_i})^2 > 0$ is not consumed by the mirror-side axioms [3, Definitions 4.11, 4.21 and 5.1]: no clause of those definitions evaluates $(A|_{V_i})^2$. Moreover, there is a numerical central-fibre model*

$$V_1 = \mathrm{Bl}_7\,\mathbb{P}^2, \qquad V_2 = \mathrm{Bl}_{11}\,\mathbb{P}^2,$$

with distinct blown-up points on a smooth plane cubic, with $q = 2$, candidate monodromy matrices

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

under Assumption 4.9(1) and Remark 4.12 of [3], Euler numbers 10 and 14, a nef restriction class on V_1 of square 0, and also a nef restriction class on V_1 of positive square. Thus the corrected clause is non-vacuous as a condition on A, but it is redundant for the lattice and monodromy clauses of the mirror-side axioms. No global family, global divisor A, very-general occurrence, or full satisfaction of the mirror axioms is asserted.

Proof. First inspect the definitions. Definition 4.11 declares a loop allowable precisely when Assumption 4.9 holds for both disc fibrations. Assumption 4.9(1) is a monodromy-matrix condition, and Assumption 4.9(2) is the primitivity of $r(a)$ in relative homology. Definition 4.21 is a lattice-incidence condition: the quasipolarisation lattice contains the fibre class F, and its orthogonal complement contains the lifted class τ. Definition 5.1 consists of primitive-embedding conditions, the existence of the isometry ψ, the lift $\widehat{\psi}$, and m-admissibility. None of these clauses contains or evaluates the number $(A|_{V_i})^2$.

Now construct the numerical witness. Let

$$V_d = \mathrm{Bl}_d\,\mathbb{P}^2$$

with the blown-up points distinct and lying on a smooth cubic, and let D be the strict transform of the cubic. As in Proposition 5.2,

$$K_{V_d}^2 = D^2 = 9 - d, \qquad \mathrm{e}(V_d) = 3 + d.$$

For $d_1 = 7$ and $d_2 = 11$ one has $d_1 + d_2 = 18$, hence the numerical d-semistability condition holds. The invariants are

$$K_{V_1}^2 = 2, \qquad K_{V_2}^2 = -2,$$

and

$$\mathrm{e}(V_1) = 10, \qquad \mathrm{e}(V_2) = 14.$$

Thus $q = K_{V_1}^2 = 2$, and, if Assumption 4.9(1) and Remark 4.12 of [3] are imposed, the corresponding inverse monodromy matrices are

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

On $V_1 = \mathrm{Bl}_7 \mathbb{P}^2$, let h be the pullback of the hyperplane class and E_1 the exceptional divisor over the first blown-up point. Put

$$P = h - E_1.$$

The complete linear system $|h - E_1|$ is the proper transform of the pencil of lines through the first blown-up point. On $\mathrm{Bl}_{p_1} \mathbb{P}^2$ it is base-point-free; after blowing up the remaining distinct points it remains base-point-free, because for every later blown-up point there is a line through p_1 not passing through that point. Hence P is nef. Its self-intersection is

$$P^2 = (h - E_1)^2 = h^2 - 2h \cdot E_1 + E_1^2 = 1 - 0 - 1 = 0.$$

Therefore P is nef but not big: a big nef divisor on a smooth projective surface has positive self-intersection. Taking $A|_{V_1} = P$ gives a central-fibre restriction with square 0, violating the corrected positivity clause on V_1, while the numerical monodromy data remain consistent with $q = 2$.

On the other hand, the hyperplane class h on $V_1 = \mathrm{Bl}_7 \mathbb{P}^2$ is nef and satisfies

$$h^2 = 1 > 0.$$

Thus the corrected clause is not vacuous: it excludes classes such as $P = h - E_1$ of square zero, while admitting classes such as h of positive square.

This proves that the corrected clause excludes some central-fibre restriction classes, so it is non-vacuous as a condition on A, but also that no clause of Definitions 4.11, 4.21 or 5.1 uses this inequality. The construction is only a restriction-class and numerical-monodromy witness: it does not construct a global effective flat nef Cartier divisor A on a total family, nor does it establish very-general occurrence or the full mirror axioms. □

7 Open points

The following points remain open.

1. The original existential assertion is not proved in this paper: for a very general H-polarised Tyurin degeneration and a very general $H \oplus E_8 \oplus E_8$-quasipolarised K3 surface, no compatible elliptic fibration π and allowable loop γ are constructed.
2. Assumption 4.9(2), the primitivity of $r(a)$ in the relative homology of the disc fibration, remains an explicit hypothesis in our monodromy analysis.
3. The relation between the corrected positivity clause $(A|_{V_i})^2 > 0$ and the non-balanced condition $q \neq 0$ is not proved. In this paper $q \neq 0$ is used only as a standing hypothesis.
4. The balanced case $q = 0$ is not settled under the problem hypotheses.
5. For an elliptic fibration with singular fibres $2\,\mathrm{II}^* + 4\,\mathrm{I}_1$, the residual cases $|q| \in \{0, 1, 8, 9\}$ require the actual monodromy representation and vanishing-cycle conjugation data.
6. The existence, uniqueness and classification of elliptic fibrations on the very general $H \oplus E_8 \oplus E_8$-quasipolarised K3 surface, including the possible occurrence of non-extremal $24\,\mathrm{I}_1$ models, remain open.
7. The precise cusp-complement type $\Gamma \cong E_8 \oplus E_8$ versus $\Gamma \cong D_{16}^+$, and its effect on the $\psi(\Gamma) = (\operatorname{im} L)^\perp$ conjunct for a specific degeneration, is not resolved.
8. The representability, closedness and global stack-theoretic structure of the mirror-pair incidence locus are open.
9. The precise boundary-stratum interpretation of the very-general Noether–Lefschetz condition in the Alexeev–Brunyate–Engel compactification needs further clarification.
10. The global existence of the divisor A in the positivity-clause witness is not established.

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